



Southeast University

Department of Computer Science & Engineering

Computer Networking Project

CEP

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Section – 19

Course Code – CSE342

Course Title – Computer Networking Lab

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1. Objective

The objectives of this project are to:

1. Design a VLSM-based IPv4 addressing scheme for a six-branch enterprise network from the single assigned block 192.168.100.0/23, allocating only as many addresses as each network needs.
2. Automate host configuration in every branch LAN with DHCP, so that end devices receive their IP address, subnet mask, default gateway and DNS server without manual entry.
3. Deploy two DNS servers and six web servers (three per DNS server) across two branch sites, and confirm that each domain name resolves to the correct web server.
4. Configure OSPF across the six-router backbone so that every branch can reach every other branch dynamically, without static routes.
5. Build a two-switch, three-VLAN network with access and trunk ports that keeps the three device groups isolated at Layer 2.
6. Implement NAT-PAT on an edge router so that many private hosts share a small pool of public addresses when communicating with an outside network.

2. Introduction

This project designs and builds the network of a multi-branch enterprise in Cisco Packet Tracer. Six branch routers are connected in a chain, and each branch has a LAN sized for a different number of hosts (35, 25, 16, 12, 8 and 4). Every subnet in the design is carved from one assigned block, **192.168.100.0/23**, using Variable Length Subnet Masking (VLSM) so that small LANs do not waste large address blocks.

Four services are layered on top of the addressing plan. DHCP gives every LAN automatic host configuration. DNS servers paired with web servers provide name-based access to six internal websites. OSPF lets the six routers learn each other's networks automatically. A stand-alone two-switch segment demonstrates VLAN isolation. Separately, a pair of edge routers demonstrates NAT-PAT, translating a private LAN to public addresses.

The project follows the three canvases of the supplied model file, **Base Setup**, **VLAN Setup** and **NAT-PAT Setup**, shown as Figures 1 to 3 in Section 3.1. The inputs assigned to Group 9 are listed below.

Table 1 - Group 9 assigned inputs

Item	Value
Group / members	Group 9 - 2024100000429, 2024100000437
Base IPv4 block	192.168.100.0/23 (192.168.100.0 - 192.168.101.255, 512 addresses)
Core LAN host sizes	Router 1: 35, Router 2: 25, Router 3: 16, Router 4: 12, Router 5: 8, Router 6: 4
Router-to-router link size	4 addresses (/30)
VLANs (host size)	VLAN 101 (30), VLAN 102 (15), VLAN 103 (10)
NAT router LAN size	15 hosts

3. Implementation

3.1 Network Topology

The topology is taken directly from the supplied model project file. Router0 to Router5 form the six-branch backbone, each with a LAN switch, a DHCP server and PCs. The first branch (Router0) and the

last branch (Router5) also host a DNS server and three web servers each. Figure 2 is a self-contained two-switch VLAN segment and Figure 3 is the NAT-PAT pair, Router6 (inside) and Router7 (outside).

The three canvases are not cabled to one another in the model file, and each demonstrates one requirement independently.

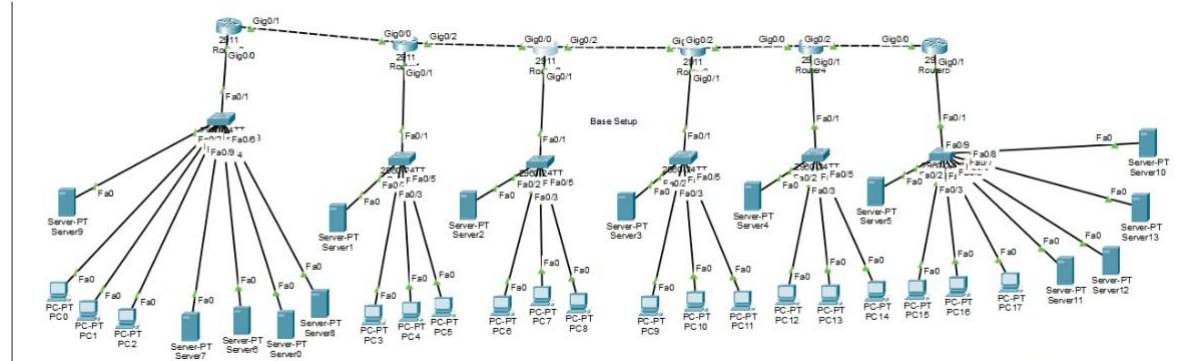


Figure 1 - Base Setup: Router0 to Router5 in a chain, each with its own switch, DHCP server and PCs. Router0 and Router5 carry the DNS and web servers.

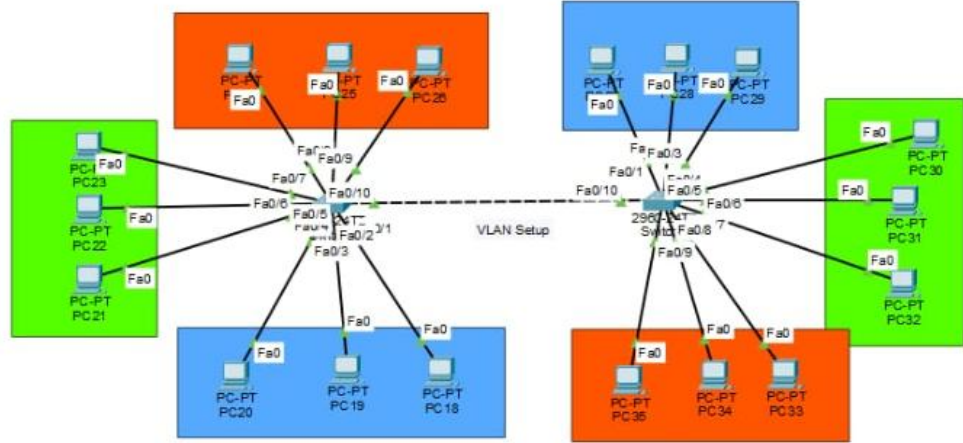


Figure 2 - VLAN Setup: two switches trunked together, each carrying all three VLANs (blue, green, orange), with no router.

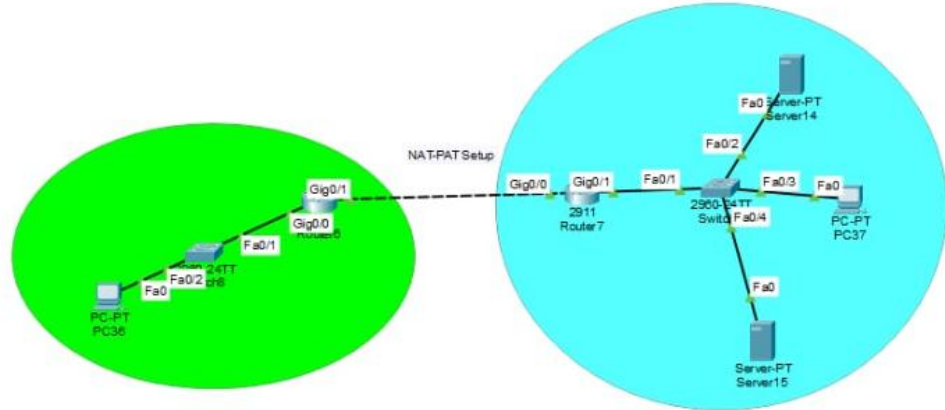


Figure 3 - NAT-PAT Setup: Router6 (private inside LAN with PC36) connected to Router7 (outside LAN with Server14, Server15 and PC37).

Table 2 - Device roles used in this report (Figure 1 and Figure 3)

Packet Tracer device	Role	Location
Router0 ... Router5	Branch routers 1-6	Backbone chain (Figure 1)

Packet Tracer device	Role	Location
Server9	DHCP Server 1	Router0 LAN
Server0	DNS Server 1	Router0 LAN
Server6, Server7, Server8	Web Server 1, 2, 3	Router0 LAN
Server1, Server2, Server3, Server4	DHCP Server 2, 3, 4, 5	Router1, Router2, Router3, Router4 LANs
Server5	DHCP Server 6	Router5 LAN
Server10	DNS Server 2	Router5 LAN
Server11, Server12, Server13	Web Server 4, 5, 6	Router5 LAN
Router6 / Router7	NAT inside router / outside router	NAT-PAT canvas (Figure 3)
Server14, Server15, PC37	Outside (simulated internet) hosts	Behind Router7

Assumptions

1. The Base Setup is a chain of six routers, which gives **five** router-to-router links. A sixth /30 is reserved in the addressing plan in case a Router5-Router0 link is required (see Section 4).
2. Each LAN has its own dedicated DHCP server (Server-PT device), not the router's built-in DHCP service. Only the NAT LAN, which has no server, uses router-based DHCP.
3. The model shows three PCs per LAN for demonstration. The subnets are nevertheless sized for the full host counts assigned to Group 9, plus the router, DHCP server and any DNS/web servers on that LAN.
4. The two DNS/web clusters sit on the Router0 LAN and the Router5 LAN, which is why those two LANs carry a few extra static addresses.
5. The VLAN segment has no router. PCs in it are configured statically, and devices can only communicate inside their own VLAN.
6. The NAT inside LAN uses a block from the assigned 192.168.100.0/23 space. The outside side of the NAT boundary uses 198.51.100.0/24, the IETF documentation range (RFC 5737), to represent public addressing that belongs to another organisation.

3.2 VLSM and IP Addressing (Question 1)

The plan was produced in three steps: (1) list every network and the addresses it needs, including infrastructure devices, the network address and the broadcast address; (2) round each requirement up to the next power of two; (3) allocate blocks from largest to smallest, each aligned on its own block boundary, so that the whole plan is contiguous and gap-free.

Table 3 - Address requirement and block size for each network

Network	Devices to address	Needed (+2)	Block	Prefix
Branch 1 LAN (Router0)	35 PCs + gateway + DHCP + DNS + 3 web = 41	43	64	/26
Branch 2 LAN (Router1)	25 PCs + gateway + DHCP = 27	29	32	/27
Branch 3 LAN (Router2)	16 PCs + gateway + DHCP = 18	20	32	/27
VLAN 101	30 hosts (no gateway)	32	32	/27
VLAN 102	15 hosts (no gateway)	17	32	/27
NAT inside LAN (Router6)	15 hosts + gateway = 16	18	32	/27

Network	Devices to address	Needed (+2)	Block	Prefix
Branch 4 LAN (Router3)	12 PCs + gateway + DHCP = 14	16	16	/28
Branch 5 LAN (Router4)	8 PCs + gateway + DHCP = 10	12	16	/28
Branch 6 LAN (Router5)	4 PCs + gateway + DHCP + DNS + 3 web = 10	12	16	/28
VLAN 103	10 hosts (no gateway)	12	16	/28
5 router-to-router links	2 router interfaces each	4 each	4 each	/30

VLAN 102 needs 15 hosts, but a /28 has only 14 usable addresses, so it is given a /27. Branch 4 fits a /28 exactly (12 PCs + gateway + DHCP server = 14 usable addresses).

Table 4 - Complete IP addressing table (Base Setup, VLAN Setup and NAT inside LAN)

Network	Network address	Subnet mask	Broadcast	Usable host range	Default gateway
Branch 1 LAN (Router0)	192.168.100.0	255.255.255.192 (/26)	192.168.100.63	192.168.100.1 - 192.168.100.62	192.168.100.1
Branch 2 LAN (Router1)	192.168.100.64	255.255.255.224 (/27)	192.168.100.95	192.168.100.65 - 192.168.100.94	192.168.100.65
Branch 3 LAN (Router2)	192.168.100.96	255.255.255.224 (/27)	192.168.100.127	192.168.100.97 - 192.168.100.126	192.168.100.97
VLAN 101 (blue)	192.168.100.128	255.255.255.224 (/27)	192.168.100.159	192.168.100.129 - 192.168.100.158	None
VLAN 102 (green)	192.168.100.160	255.255.255.224 (/27)	192.168.100.191	192.168.100.161 - 192.168.100.190	None
NAT inside LAN (Router6)	192.168.100.192	255.255.255.224 (/27)	192.168.100.223	192.168.100.193 - 192.168.100.222	192.168.100.193
Branch 4 LAN (Router3)	192.168.100.224	255.255.255.240 (/28)	192.168.100.239	192.168.100.225 - 192.168.100.238	192.168.100.225
Branch 5 LAN (Router4)	192.168.100.240	255.255.255.240 (/28)	192.168.100.255	192.168.100.241 - 192.168.100.254	192.168.100.241
Branch 6 LAN (Router5)	192.168.101.0	255.255.255.240 (/28)	192.168.101.15	192.168.101.1 - 192.168.101.14	192.168.101.1
VLAN 103 (orange)	192.168.101.16	255.255.255.240 (/28)	192.168.101.31	192.168.101.17 - 192.168.101.30	None
Link Router0-Router1	192.168.101.32	255.255.255.252 (/30)	192.168.101.35	192.168.101.33 - 192.168.101.34	None
Link Router1-Router2	192.168.101.36	255.255.255.252 (/30)	192.168.101.39	192.168.101.37 - 192.168.101.38	None
Link Router2-Router3	192.168.101.40	255.255.255.252 (/30)	192.168.101.43	192.168.101.41 - 192.168.101.42	None
Link Router3-Router4	192.168.101.44	255.255.255.252 (/30)	192.168.101.47	192.168.101.45 - 192.168.101.46	None
Link Router4-Router5	192.168.101.48	255.255.255.252 (/30)	192.168.101.51	192.168.101.49 - 192.168.101.50	None
Reserved: Router5-Router0	192.168.101.52	255.255.255.252 (/30)	192.168.101.55	192.168.101.53 - 192.168.101.54	None

Ten LAN/VLAN networks and five backbone links use 308 of the 512 addresses in 192.168.100.0/23 (about 60%). The reserved /30 takes 4 more, and 200 addresses (192.168.101.56 - 192.168.101.255) remain free for future growth.

Table 5 - Router interface addressing (Figure 1)

Router	Interface	IP address	Subnet mask	Connects to
Router0	Gig0/0	192.168.100.1	255.255.255.192	Branch 1 LAN

Router	Interface	IP address	Subnet mask	Connects to
Router0	Gig0/1	192.168.101.33	255.255.255.252	Router1 Gig0/0
Router1	Gig0/0	192.168.101.34	255.255.255.252	Router0 Gig0/1
Router1	Gig0/1	192.168.100.65	255.255.255.224	Branch 2 LAN
Router1	Gig0/2	192.168.101.37	255.255.255.252	Router2 Gig0/0
Router2	Gig0/0	192.168.101.38	255.255.255.252	Router1 Gig0/2
Router2	Gig0/1	192.168.100.97	255.255.255.224	Branch 3 LAN
Router2	Gig0/2	192.168.101.41	255.255.255.252	Router3 Gig0/0
Router3	Gig0/0	192.168.101.42	255.255.255.252	Router2 Gig0/2
Router3	Gig0/1	192.168.100.225	255.255.255.240	Branch 4 LAN
Router3	Gig0/2	192.168.101.45	255.255.255.252	Router4 Gig0/0
Router4	Gig0/0	192.168.101.46	255.255.255.252	Router3 Gig0/2
Router4	Gig0/1	192.168.100.241	255.255.255.240	Branch 5 LAN
Router4	Gig0/2	192.168.101.49	255.255.255.252	Router5 Gig0/0
Router5	Gig0/0	192.168.101.50	255.255.255.252	Router4 Gig0/2
Router5	Gig0/1	192.168.101.1	255.255.255.240	Branch 6 LAN

Table 6 - Static server addresses and DHCP pools (excluded from dynamic assignment)

LAN	Static devices	Dynamic pool for PCs
Branch 1	Router0 gateway 192.168.100.1, DHCP Server 1 192.168.100.2, DNS Server 1 192.168.100.3, Web Server 1/2/3 192.168.100.4 / .5 / .6	192.168.100.7 - 192.168.100.62 (56 addresses)
Branch 2	Router1 gateway 192.168.100.65, DHCP Server 2 192.168.100.66	192.168.100.67 - 192.168.100.94 (28)
Branch 3	Router2 gateway 192.168.100.97, DHCP Server 3 192.168.100.98	192.168.100.99 - 192.168.100.126 (28)
Branch 4	Router3 gateway 192.168.100.225, DHCP Server 4 192.168.100.226	192.168.100.227 - 192.168.100.238 (12)
Branch 5	Router4 gateway 192.168.100.241, DHCP Server 5 192.168.100.242	192.168.100.243 - 192.168.100.254 (12)
Branch 6	Router5 gateway 192.168.101.1, DHCP Server 6 192.168.101.2, DNS Server 2 192.168.101.3, Web Server 4/5/6 192.168.101.4 / .5 / .6	192.168.101.7 - 192.168.101.14 (8)

Router interface configuration

The interface addresses above are applied with the commands below. The complete per-router script, including OSPF, is given in Section 3.5.

```
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.100.1 255.255.255.192
Router0(config-if)# no shutdown
Router0(config-if)# interface GigabitEthernet0/1
Router0(config-if)# ip address 192.168.101.33 255.255.255.252
Router0(config-if)# no shutdown
```

3.3 DHCP Configuration (Question 2)

Every LAN in Figure 1 has its own Server-PT acting as the DHCP server, so DHCP is configured on the servers rather than with ip dhcp pool on the routers. Each pool hands out the gateway of its own

LAN and a DNS server address. Branches 1 to 3 use DNS Server 1 and Branches 4 to 6 use DNS Server 2, since each is the nearer DNS server along the chain.

Table 7 - DHCP server settings for each branch LAN

Branch	DHCP server (static IP)	Default gateway	DNS server	Start IP	Subnet mask	Max users
1	192.168.100.2	192.168.100.1	192.168.100.3	192.168.100.7	255.255.255.192	56
2	192.168.100.66	192.168.100.65	192.168.100.3	192.168.100.67	255.255.255.224	28
3	192.168.100.98	192.168.100.97	192.168.100.3	192.168.100.99	255.255.255.224	28
4	192.168.100.226	192.168.100.225	192.168.101.3	192.168.100.227	255.255.255.240	12
5	192.168.100.242	192.168.100.241	192.168.101.3	192.168.100.243	255.255.255.240	12
6	192.168.101.2	192.168.101.1	192.168.101.3	192.168.101.7	255.255.255.240	8

Procedure on each DHCP server: open the server, go to Desktop > IP Configuration and set the static IP, mask and gateway from Table 7. Then open Services > DHCP, turn the service On, and fill in Default Gateway, DNS Server, Start IP Address, Subnet Mask and Maximum Number of Users exactly as listed, and click Save. The pool is named serverPool by default and may be left as is.

On each PC: open Desktop > IP Configuration and select DHCP. The PC then shows the address, mask, gateway and DNS server given in the table.

The NAT inside LAN (Figure 3) has no server, so Router6 provides DHCP itself. This is configured together with NAT in Section 3.7.

Verification

Test	Expected result
PC0 (Branch 1) set to DHCP	Address in 192.168.100.7 - 192.168.100.62, mask 255.255.255.192, gateway 192.168.100.1, DNS 192.168.100.3
PC3 (Branch 2) set to DHCP	Address in 192.168.100.67 - 192.168.100.94, gateway 192.168.100.65, DNS 192.168.100.3
PC9 (Branch 4) set to DHCP	Address in 192.168.100.227 - 192.168.100.238, gateway 192.168.100.225, DNS 192.168.101.3
PC15 (Branch 6) set to DHCP	Address in 192.168.101.7 - 192.168.101.14, gateway 192.168.101.1, DNS 192.168.101.3
ipconfig /all on any PC	Shows DHCP server address equal to that branch's DHCP server

3.4 DNS and Web Server Deployment (Question 3)

Two DNS servers and six web servers are deployed in two clusters: DNS Server 1 with Web Servers 1 to 3 on the Router0 LAN, and DNS Server 2 with Web Servers 4 to 6 on the Router5 LAN. Each web server hosts one distinct site.

Table 8 - DNS and web server deployment

Server	IP address	Hosted website	Cluster
DNS Server 1	192.168.100.3	-	Router0 LAN
Web Server 1	192.168.100.4	www.seu.edu.bd	Router0 LAN
Web Server 2	192.168.100.5	www.prothomalo.com	Router0 LAN
Web Server 3	192.168.100.6	www.daraz.com.bd	Router0 LAN
DNS Server 2	192.168.101.3	-	Router5 LAN
Web Server 4	192.168.101.4	www.bkash.com	Router5 LAN

Server	IP address	Hosted website	Cluster
Web Server 5	192.168.101.5	www.dhakatribune.com	Router5 LAN
Web Server 6	192.168.101.6	www.buet.ac.bd	Router5 LAN

DNS servers: set the static IP first, then open Services > DNS, turn the service On, and add an A record for each of the six sites (Name = domain, Address = the web server's IP). Both DNS servers hold all six A records. This is deliberate: Branches 1-3 query DNS Server 1 and Branches 4-6 query DNS Server 2, so holding the full set on both means a PC in any branch can open any of the six sites.

Web servers: set the static IP, mask and gateway, then open Services > HTTP and edit index.html so that each of the six pages shows a different heading naming its own site, making it obvious which server answered.

Verification

Test	Expected result
PC in Branch 2: browser to www.seu.edu.bd	Web Server 1 page loads (DNS Server 1 resolves, HTTP succeeds)
PC in Branch 5: browser to www.bkash.com	Web Server 4 page loads (DNS Server 2 resolves, crossing the OSPF backbone)
PC in Branch 1: browser to www.buet.ac.bd	Web Server 6 page loads, proving both DNS servers hold all records
nslookup www.daraz.com.bd from a PC	Returns 192.168.100.6

3.5 OSPF Dynamic Routing (Question 4)

OSPF process 1 runs in a single area (area 0) on Router0 to Router5. Each router advertises its own LAN and its one or two backbone links, using wildcard masks that match the subnet sizes in Table 4. The LAN-facing interface of each router is set as passive, so hello packets are not sent toward PCs and servers while the LAN network is still advertised. Router6 and Router7 stay outside the OSPF domain because they belong to the NAT canvas.

Router0 and Router5 are the two ends of the chain, so each has a single backbone link. Router1 to Router4 each have two.

Router0

```
enable
configure terminal
interface GigabitEthernet0/0
description Branch 1 LAN gateway
ip address 192.168.100.1 255.255.255.192
no shutdown
exit
interface GigabitEthernet0/1
description link to Router1
ip address 192.168.101.33 255.255.255.252
no shutdown
exit
router ospf 1
router-id 1.1.1.1
passive-interface GigabitEthernet0/0
network 192.168.100.0 0.0.0.63 area 0
network 192.168.101.32 0.0.0.3 area 0
exit
end
write memory
```

Router1

```
enable
configure terminal
interface GigabitEthernet0/1
description Branch 2 LAN gateway
ip address 192.168.100.65 255.255.255.224
no shutdown
exit
interface GigabitEthernet0/0
description link to Router0
ip address 192.168.101.34 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/2
description link to Router2
ip address 192.168.101.37 255.255.255.252
no shutdown
exit
router ospf 1
router-id 2.2.2.2
passive-interface GigabitEthernet0/1
network 192.168.100.64 0.0.0.31 area 0
network 192.168.101.32 0.0.0.3 area 0
network 192.168.101.36 0.0.0.3 area 0
exit
end
write memory
```

Router2

```
enable
configure terminal
interface GigabitEthernet0/1
description Branch 3 LAN gateway
ip address 192.168.100.97 255.255.255.224
no shutdown
exit
interface GigabitEthernet0/0
description link to Router1
ip address 192.168.101.38 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/2
description link to Router3
ip address 192.168.101.41 255.255.255.252
no shutdown
exit
router ospf 1
router-id 3.3.3.3
passive-interface GigabitEthernet0/1
network 192.168.100.96 0.0.0.31 area 0
network 192.168.101.36 0.0.0.3 area 0
network 192.168.101.40 0.0.0.3 area 0
exit
end
write memory
```

Router3

```
enable
configure terminal
interface GigabitEthernet0/1
description Branch 4 LAN gateway
ip address 192.168.100.225 255.255.255.240
no shutdown
exit
interface GigabitEthernet0/0
description link to Router2
ip address 192.168.101.42 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/2
description link to Router4
ip address 192.168.101.45 255.255.255.252
no shutdown
exit
router ospf 1
router-id 4.4.4.4
passive-interface GigabitEthernet0/1
network 192.168.100.224 0.0.0.15 area 0
network 192.168.101.40 0.0.0.3 area 0
network 192.168.101.44 0.0.0.3 area 0
exit
end
write memory
```

Router4

```
enable
configure terminal
interface GigabitEthernet0/1
description Branch 5 LAN gateway
ip address 192.168.100.241 255.255.255.240
no shutdown
exit
interface GigabitEthernet0/0
description link to Router3
ip address 192.168.101.46 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/2
description link to Router5
ip address 192.168.101.49 255.255.255.252
no shutdown
exit
router ospf 1
router-id 5.5.5.5
passive-interface GigabitEthernet0/1
network 192.168.100.240 0.0.0.15 area 0
network 192.168.101.44 0.0.0.3 area 0
network 192.168.101.48 0.0.0.3 area 0
exit
end
write memory
```

Router5

```
enable
configure terminal
interface GigabitEthernet0/1
description Branch 6 LAN gateway
ip address 192.168.101.1 255.255.255.240
no shutdown
exit
interface GigabitEthernet0/0
description link to Router4
ip address 192.168.101.50 255.255.255.252
no shutdown
exit
router ospf 1
router-id 6.6.6.6
passive-interface GigabitEthernet0/1
network 192.168.101.0 0.0.0.15 area 0
network 192.168.101.48 0.0.0.3 area 0
exit
end
write memory
```

Verification

Command / test	Expected result
show ip ospf neighbor (Router2)	Two neighbours in FULL state: Router1 and Router3
show ip route ospf (Router0)	O routes for every other LAN and every link not directly attached
show ip route (Router5)	Routes to 192.168.100.0/26 through 192.168.101.50
Ping from a Branch 1 PC to a Branch 6 PC	Replies (crosses all five links)
Ping from a Branch 3 PC to a Branch 4 PC	Replies
tracert from a Branch 1 PC to a Branch 6 PC	Shows each router in the chain in order

3.6 VLAN Architecture and Traffic Isolation (Question 5)

Figure 2 contains two switches, each with nine access ports in use and one trunk port. Both switches carry all three VLANs. VLAN 101 is the blue group, VLAN 102 the green group and VLAN 103 the orange group. On each switch, three access ports serve each colour, and the trunk to the other switch is on Fa0/10. There is no router or Layer 3 device in this segment.

Table 9 - VLAN plan and PC addressing (static, no gateway)

VLAN	Subnet	Switch 6 PCs	Switch 7 PCs
VLAN 101 - blue	192.168.100.128/27 mask 255.255.255.224	PC18 192.168.100.129 PC19 192.168.100.130 PC20 192.168.100.131	PC27 192.168.100.132 PC28 192.168.100.133 PC29 192.168.100.134
VLAN 102 - green	192.168.100.160/27 mask 255.255.255.224	PC21 192.168.100.161 PC22 192.168.100.162 PC23 192.168.100.163	PC30 192.168.100.164 PC31 192.168.100.165 PC32 192.168.100.166
VLAN 103 - orange	192.168.101.16/28 mask 255.255.255.240	PC24 192.168.101.17 PC25 192.168.101.18 PC26 192.168.101.19	PC33 192.168.101.20 PC34 192.168.101.21 PC35 192.168.101.22

Switch A is the switch on the left of Figure 2 (PC18 to PC26) and Switch B is the one on the right (PC27 to PC35). Because there is no router or DHCP server on this canvas, the PCs are configured manually under Desktop > IP Configuration, and the default gateway is left blank.

Switch 6 configuration (Switch 7 is identical)

```
enable
configure terminal
hostname VLAN-SW1      ! use VLAN-SW2 on the second switch
vlan 101
  name BLUE
vlan 102
  name GREEN
vlan 103
  name ORANGE
exit
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 101
interface range FastEthernet0/4 - 6
  switchport mode access
  switchport access vlan 102
interface range FastEthernet0/7 - 9
  switchport mode access
  switchport access vlan 103
interface FastEthernet0/10
  switchport mode trunk
  switchport trunk allowed vlan 101,102,103
exit
end
write memory
```

Verification

Test	Expected result
show vlan brief (each switch)	VLAN 101, 102, 103 exist with three access ports each
show interfaces trunk (each switch)	Fa0/10 is trunking and allows VLANs 101, 102, 103
PC18 pings PC27 (VLAN 101, across the trunk)	Replies
PC21 pings PC30 (VLAN 102, across the trunk)	Replies
PC24 pings PC33 (VLAN 103, across the trunk)	Replies
PC18 pings PC21 (VLAN 101 to 102)	Request timed out
PC21 pings PC24 (VLAN 102 to 103)	Request timed out

3.7 NAT-PAT Edge Configuration (Question 6)

In Figure 3, Router6 is the NAT router. Its Gig0/0 faces the private inside LAN (Switch and PC36) and is marked ip nat inside. Its Gig0/1 faces Router7 and is marked ip nat outside. Router7 and its LAN (Server14, Server15, PC37) represent the outside world and do not run NAT.

Instead of translating to a single interface address, Router6 translates the whole inside LAN into a **pool of two public addresses** with the overload keyword. This is many-to-few PAT: hosts share the pool addresses and are told apart by source port.

Table 10 - NAT-PAT addressing

Side	Device / interface	IP address	Role
Private (inside)	Router6 Gig0/0	192.168.100.193 / 255.255.255.224	Inside gateway (ip nat inside)
Private (inside)	PC36 and other inside hosts	192.168.100.194 - 192.168.100.222 (DHCP)	Translated when leaving
Transit link	Router6 Gig0/1	198.51.100.1 / 255.255.255.248	Outside interface (ip nat outside)
Transit link	Router7 Gig0/0	198.51.100.2 / 255.255.255.248	Router7 side of the link
Outside LAN	Router7 Gig0/1	198.51.100.9 / 255.255.255.248	Gateway for outside hosts
Outside LAN	Server14	198.51.100.10 / 255.255.255.248	Outside host
Outside LAN	Server15	198.51.100.11 / 255.255.255.248	Outside host
Outside LAN	PC37	198.51.100.12 / 255.255.255.248	Outside host
Public NAT pool	PUBLIC-POOL	198.51.100.17 - 198.51.100.18 (/29)	Addresses the inside LAN is translated to

Outside hosts (Server14, Server15, PC37) use gateway 198.51.100.9 and mask 255.255.255.248.

Router6 (NAT router)

```
enable
configure terminal
interface GigabitEthernet0/0
description Inside private LAN
ip address 192.168.100.193 255.255.255.224
ip nat inside
no shutdown
interface GigabitEthernet0/1
description Outside link to Router7
ip address 198.51.100.1 255.255.255.248
ip nat outside
no shutdown
exit
ip dhcp excluded-address 192.168.100.193
ip dhcp pool NAT-LAN
network 192.168.100.192 255.255.255.224
default-router 192.168.100.193
exit
access-list 1 permit 192.168.100.192 0.0.0.31
ip nat pool PUBLIC-POOL 198.51.100.17 198.51.100.18 netmask 255.255.255.248
ip nat inside source list 1 pool PUBLIC-POOL overload
ip route 198.51.100.8 255.255.255.248 198.51.100.2
end
write memory
```

Router7 (outside router, no NAT)

```
enable
configure terminal
interface GigabitEthernet0/0
ip address 198.51.100.2 255.255.255.248
no shutdown
interface GigabitEthernet0/1
ip address 198.51.100.9 255.255.255.248
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 198.51.100.1
end
write memory
```

The overload keyword is what makes this PAT. Without it, each inside host would need its own public address. The static route on Router6 lets it reach the outside LAN, and the default route on Router7 sends return traffic for the pool addresses back to Router6, where it is translated back to the private host.

Verification

Test	Expected result
PC36 set to DHCP	Gets an address in 192.168.100.194 - 192.168.100.222 with gateway 192.168.100.193
PC36 pings Server14 (198.51.100.10)	Replies
show ip nat translations (Router6)	Inside local 192.168.100.x mapped to inside global 198.51.100.17 or .18, with port numbers
show ip nat statistics (Router6)	Shows active translations and hit count
PC37 pings PC36's private address	Fails, because private addresses are not reachable from outside
Simulation mode: PC36 to Server14	Source address changes from 192.168.100.x to 198.51.100.17 as the packet leaves Router6

4. Discussion

The network was designed to follow the supplied model project file. One difference exists between the written requirement and the model. The project statement mentions six router-to-router links, while the model connects Router0 to Router5 in a line, which gives five point-to-point links. The chain still provides a complete path between all six routers, and OSPF makes every LAN reachable from every other. To keep the design flexible, a sixth /30 (192.168.101.52/30) is reserved, so a Router5-Router0 link could be added to close the chain into a ring without changing any existing subnet.

The VLSM plan sized every subnet against its actual needs, including the router, DHCP server and any DNS or web servers. This is why Branch 1 and Branch 6 carry a few more addresses than the bare PC counts, and why VLAN 102 uses a /27 rather than a /28. Allocating largest-first with each block aligned to its own size kept the plan contiguous, with the whole 192.168.101.56 - 192.168.101.255 range left free.

DNS is split across two servers, each placed closest to the branches that use it, which keeps DNS traffic short in the normal case. Both servers hold the full set of six A records, so a lookup succeeds whichever branch it comes from. Using a single DNS server for everything would also work because OSPF provides full reachability, so the two-server layout is a design choice that also satisfies the requirement for two DNS servers.

The VLAN part was kept as a pure Layer 2 network. Devices in the same VLAN communicate across the trunk between the two switches, while different VLANs cannot communicate because no router or Layer 3 switch exists to route between them. Adding router-on-a-stick would create inter-VLAN traffic, so it was intentionally left out. The NAT part uses a pool with overload, so more than one public address is used, and the private 192.168.100.192/27 network never appears outside Router6. The outside side uses the RFC 5737 documentation range because it must not clash with real internet addresses.

Each service was checked with a specific test rather than assumed to work: DHCP leases on PCs, website loading by domain name, OSPF neighbours and routes followed by cross-LAN pings, same-VLAN versus cross-VLAN pings, and the NAT translation table. The expected results are listed under each question in Section 3, and the same tests are used during the demonstration.

Limitations and possible extensions

- The single-area chain has no redundancy: if one backbone link fails, the branches beyond it are cut off. The reserved /30 allows a ring to be added.
- DHCP and DNS are single servers per site, which are single points of failure. A second server or router-based DHCP would add resilience.
- The VLAN segment could later be attached to the backbone using router-on-a-stick and access lists to permit selected inter-VLAN traffic.
- OSPF authentication and port security on the access switches would improve security in a real deployment.

5. Conclusion

The assigned block 192.168.100.0/23 was divided with VLSM into ten LAN and VLAN networks plus five router-to-router links, using 308 of the 512 addresses and leaving a large contiguous free range plus one reserved link. On this plan the project delivers dedicated DHCP servers on every branch LAN, two DNS servers each backed by three distinct websites, single-area OSPF across the six-router chain, a router-free two-switch VLAN segment that isolates three groups at Layer 2, and a NAT-PAT edge that translates a private LAN to a small public pool. Each of the six project questions maps onto a part of the supplied model topology and has a defined test that confirms it works.

6. References

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